

Encounter-Level Neural Surrogate Workstream

Comprehensive Mentor Update: Motivation, Data Generation, Experiments, Failures, Final Ablation, and Conclusions

Prepared from the encounterNN discussion and uploaded run outputs

Headline conclusion: the encounterNN residual model successfully learned a difficult local close-encounter correction, but it did not outperform the existing scalar Zone-3 NN in the optimized full-pipeline ablation. It is best treated as a strong prototype/future-work result, not as a replacement for the current production SIMON method.

Scope note: smoke tests are intentionally omitted, as requested. This report focuses on the substantive experiment sequence, generated data, trained models, diagnostics, decisions, and final evaluator results.

1. Executive Summary

The encounterNN workstream began as an attempt to move beyond the existing scalar Zone-3 force-correction approach. The goal was to test whether a neural model could learn a whole encounter-window correction: starting from a Zone-3 encounter entry state, run a short noNN window, predict the residual between IAS15 and noNN at the encounter exit, apply that residual, and resume the long SIMON rollout.

Question	Final answer
Did the encounterNN learn the local event?	Yes. On the exact IC1 event at t=82.56 yr, local position exit error improved by 77.74% and local velocity exit error improved by 95.11%.
Did it improve the full 100-year rollout?	Mixed. It improved final position, final velocity, and velocity time-average, but worsened position time-average by about 2.0% versus SIMON-noNN.
Did it beat the scalarNN production baseline?	No. SIMON-scalarNN remained better on the main position metrics and retained strong speedup.
Did it provide speedup?	No. The residual encounterNN still requires the noNN 0.5-year window, then adds NN inference. In the final evaluator it ran at 0.51x vs IAS15, while SIMON-noNN was 2.96x and SIMON-scalarNN was 2.76x.
Should it replace scalar Zone-3 NN?	No. The current production method should remain the scalar Zone-3 NN. The encounterNN should be written as a prototype/future-work result.

2. Starting Point Before the EncounterNN Work

The starting point was the existing v4 scalar Zone-3 evaluator: `pair_eval_v4_zone3_frozen_timeavg_v3_with_log.py`. This script was already a controlled and optimized evaluator for the revised Zone-2 / Zone-3 SIMON method. It used:

- Zone 2: direct Newtonian force with adaptive sub-stepping; no NN in Zone 2.
- Zone 3: bounded v4 trajectory-trained 6-input scalar NN, frozen per macro/sub-step.
- Zone 4: direct Newtonian force.
- v4 scalar model input: $[\log(r_{\text{soft}}), \log(m_i), \log(m_j), \log(dt), v_{\text{rad_norm}}, v_{\text{tan_norm}}]$.
- Comparison against an IAS15 reference over IC1, $T = 100$ yr, with time-averaged RMS as the main accuracy metric.

The corresponding scalar model was `pair_correction_nn_v4_bounded.pt`, trained using `train_pair_correction_v4.py` on `encounter_data_zone3_v3_augmented.npz`. The v4 scalar model was designed specifically to avoid the unsafe near-zero correction collapse seen in earlier versions by bounding `c_pred` in a safe interval.

Baseline before encounterNN: scalarNN was already a stable, production-like accuracy mechanism. The new encounterNN therefore had to justify itself either by improving accuracy beyond scalarNN, reducing NN call overhead, or opening a credible future skip-ahead path.

3. Why We Chose the Encounter-Level NN Path

The project discussion considered several ways to extend the existing scalar force-correction architecture. The options were roughly:

Option	Reason considered	Reason not chosen / status
Continue scalarNN only	It was already stable, simple, and aligned with the main paper methodology.	Kept as the production baseline, but it did not test a more ambitious learned encounter-map idea.
Learn force direction/vector force	Could in theory model richer local effects.	Earlier Vector NN direction work showed that even sub-degree direction errors can accumulate into unphysical branch changes/ejection; analytic direction should remain enforced.

Direct future-state predictor	Could potentially skip integration and save time.	Too aggressive at the start; would require a different target and more safeguards.
Encounter-level residual correction	More ambitious than scalarNN but still anchored to a noNN physical rollout. It keeps SIMON as the backbone and learns only the residual at the encounter exit.	Chosen path. It was safer than direct state prediction and directly targeted close-encounter errors.

The selected path was therefore not to replace the force law directly. It was to add a window-level residual correction module: run a physical noNN encounter window first, ask the NN to predict the IAS15-minus-noNN exit residual, then correct the exit state.

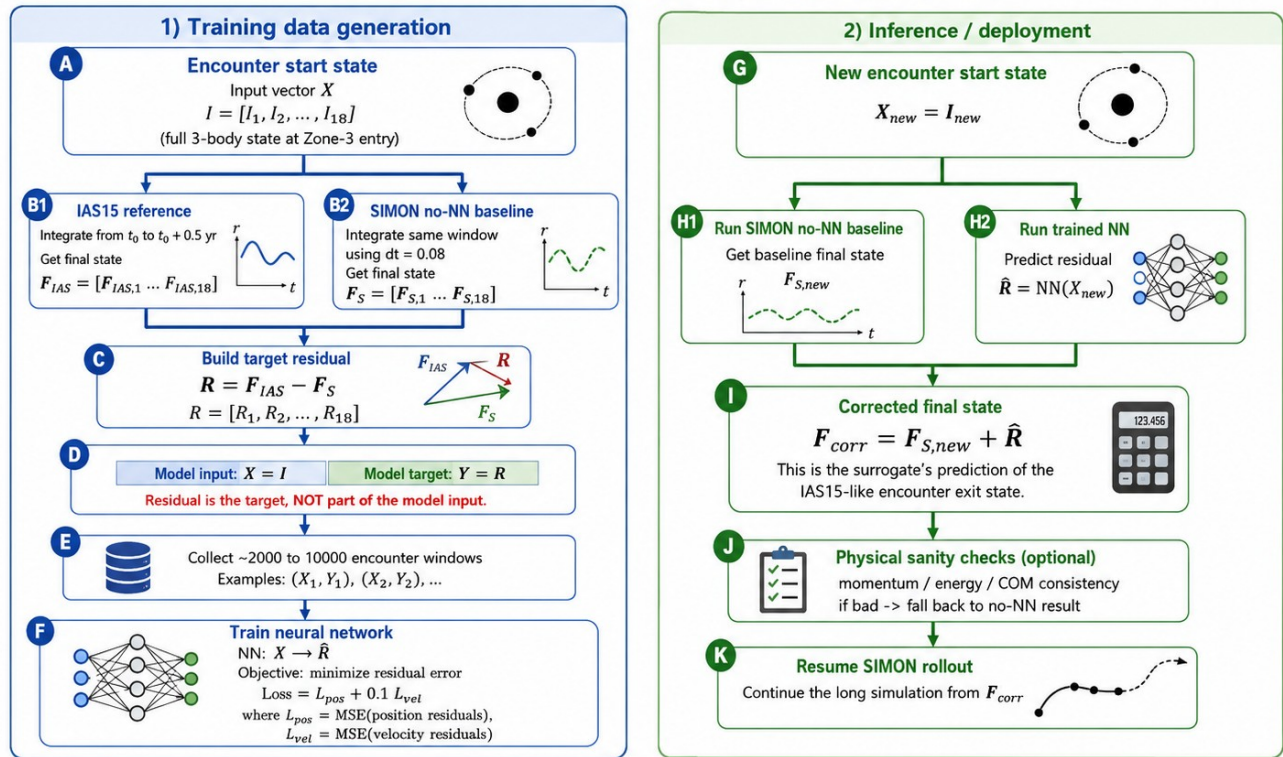
4. EncounterNN Method Definition

The encounterNN model follows the flow diagram used during the discussion. It is not a scalar force-correction model and it is not a direct IAS15-exit predictor. It is an 18-dimensional residual model.

Stage	Formula / meaning
Encounter entry	Input $X = I$ = full 3-body encounter start state, represented as 18 state variables.
IAS15 window	Integrate from t_0 to $t_0 + 0.5$ yr with IAS15 to obtain F_{IAS} .
SIMON-noNN window	Integrate the same window with SIMON-noNN to obtain F_S .
Training target	$R = F_{IAS} - F_S$.
Model	$NN(X) \rightarrow \hat{R}$.
Deployment correction	Corrected final state = $F_{S,new} + \hat{R}$.
Resume rollout	Continue long SIMON rollout from the corrected exit state.

Encounter-Level Surrogate (dt = 0.08) – Training and Inference

Goal: learn a residual correction across a 0.5-year Zone-3 encounter window



Key idea

The NN does **NOT** learn a scalar force correction c . It learns the encounter-exit residual state correction over a bounded window.



Macro-step count for 0.5 yr window at $dt = 0.08$:
6 full steps + 1 final short step (0.02 yr).

Figure 1. Encounter-level residual-correction training and inference flow used in this work.

Important distinction: because the model target is $R = F_{IAS} - F_S$, the current encounterNN still needs the

noNN window result F_S,new at inference time. Therefore, it is an accuracy-correction overlay, not a true skip-ahead model.

5. Experiment Timeline and Decisions

Phase	Main files	Data / model / outputs	Key result	Decision
1. Initial encounter data generator	generate_encounter_surrogate_data_v1.py; merge_encounter_surrogate_shards_v1.py; inspect_encounter_surrogate_data_v1.py	Initial v1 shards; later part of the pre-local 2870-sample dataset.	Basic generator worked with IAS15/noNN windows and quality filters. It produced residual-learning examples but did not guarantee coverage of the real rollout event.	Proceed to train first models, but keep strong quality filters.
2. First residual model	train_encounter_surrogate_v1.py; evaluate_encounter_surrogate_gate_v1.py	v1 residual MLP models and gated evaluation outputs.	A universal residual correction was too broad. Performance depended strongly on encounter phase.	Move toward gated deployment; do not deploy one universal model across all Zone-3 phases.
3. Velocity-safe model	train_encounter_surrogate_v2_velocitysafe.py; train_encounter_surrogate_v2_velocitysafe_fixed.py; train_encounter_surrogate_v2_velocitysafe_fixed2.py	encounter_surrogate_v2_velocitysafe_large1.pt and summary outputs.	Position improved but velocity remained unresolved or unsafe in enough cases. Code issues were fixed along the way.	Do not deploy immediately; evaluate gates and diagnose velocity/phase behavior.
4. Gate sweep and rollout prototype	evaluate_encounter_surrogate_gate_sweep_v2.py; evaluate_encounter_surrogate_fixed_gate_v2.py; evaluate_gated_encounter_surrogate_rollout_v1.py	Gate sweep outputs and rollout_surrogate_* folders.	A strong-approach gate around $v_{rad_norm} < -0.40$ emerged, but rollout initially showed no useful event or fallback behavior until the exact $t=82.56$ event was targeted.	Generate targeted data around the actual event geometry.
5. Event-targeted data	generate_encounter_surrogate_data_v2_event_targeted.py; generate_encounter_surrogate_data_v2_event_targeted_fixed.py	event-targeted shards; merged dataset reaching 2870 samples before rollout-local enrichment.	Scalar event features were well targeted: $v_{rad} \sim -1.1$, $v_{tan} \sim 0.85$, $r_{pair} \sim 0.139$ AU, $min_r \sim 0.062$ AU.	Good but not sufficient; exact rollout event still not represented well in full state-space.
6. Full-state nearest-neighbor diagnosis	diagnose_event_nearest_neighbors_v1.py; diagnose_knn_residual_surrogate_v1.py; diagnose_single_event_replay_knn_v1.py	event_nn_diag_t82p56 and KNN diagnostic folders.	The exact event was close in scalar features but not close enough in full X_{rel18} state-space. KNN/residual interpolation did not solve the mismatch.	Data coverage, not just architecture, was the main issue. Generate rollout-local data anchored to the actual trajectory.
7. Compact and hybrid models	train_encounter_surrogate_v3_compact.py; train_encounter_surrogate_v4_hybrid.py	encounter_surrogate_v3_compact_event_enriched_summary.txt; encounter_surrogate_v4_hybrid_event_enriched_summary.txt.	Changing representation/model form alone did not solve the exact event problem. The hybrid scalar idea was not sufficient to correct an 18D residual reliably.	Stop architecture-only fixes; return to data coverage.
8. Rollout-local data	generate_encounter_surrogate_data_v3_rollout_local.py	batches 202-206; encounter_surrogate_v3_rollout_local_enriched.npz; inspect_surrogate_v3_rollout_local_enriched.	Merged dataset reached 3370 samples = 2870 previous + 500 rollout-local samples. The new local samples directly covered the failed IC1 event geometry.	Train the velocity-safe full-state residual model on the rollout-local enriched dataset.
9. Final local model	train_encounter_surrogate_v2_velocitysafe_fixed2.py	encounter_surrogate_v3_rollout_local_velocitysafe.pt;	Strong held-out gains, especially on strong-approach samples.	Proceed to exact replay and full rollout.

		encounter_surrogate_v3 _rollout_local_velocitysa fe_summary.txt.	Local exact-event replay became accurate.	
10. Exact replay and alpha diagnostics	diagnose_single_event_ replay_v1.py; diagnose_rollout_error_t imeline_v1c.py	single_event_replay_v3 _rollout_local_t82p56; rollout_error_timeline_v 1c.	Exact local exit improved dramatically, but full rollout remained branch-sensitive. Partial alpha corrections did not solve the branch issue; alpha=1.0 was best of tested options.	Use full correction in final ablation, but do not claim global position improvement.
11. Final optimized ablation	pair_eval_encounterNN _ablation_v1.py	encounterNN_ablation_ v1_out/ encounterNN_ablation_ summary.txt; encounterNN_ablation_ v1_out/encounter_event s.csv.	Final ablation compared IAS15, SIMON-noNN, SIMON-scalarNN, and SIMON-encounterNN. EncounterNN improved velocity metrics but lost to scalarNN on position metrics and runtime.	Do not replace scalarNN. Present encounterNN as prototype/future work.

6. Detailed Experimental Record

6.1 Initial data-generation design

Files used / created:

- generate_encounter_surrogate_data_v1.py
- merge_encounter_surrogate_shards_v1.py
- inspect_encounter_surrogate_data_v1.py

Data, models, and outputs:

- Initial shards under encounter_surrogate_shards/
- Dataset fields included full encounter start state, noNN exit state, IAS15 exit state, residual_x/residual_v, scalar event descriptors, and quality diagnostics.

Result:

The generator was designed with quality filters: finite-state checks, IAS15 stability/ejection checks, IAS15 minimum-distance filters, IAS15 energy drift filter, noNN energy filter, and residual-outlier filters.

Discussion:

This was the right foundation because the task was not simple supervised learning from arbitrary states. The model needed physically valid encounter windows in which IAS15 remained stable and noNN was sufficiently sane to serve as a baseline.

Decision: Proceed with residual training, but gate deployment by encounter phase rather than assuming one universal Zone-3 residual model.

6.2 First residual model and gated thinking

Files used / created:

- train_encounter_surrogate_v1.py
- evaluate_encounter_surrogate_gate_v1.py

Data, models, and outputs:

- First residual MLP models trained on initial encounter-window data.
- Early gated evaluation reports.

Result:

The first model showed that a residual map could be learned in principle, but the behavior differed strongly across approaching, weak/side, and receding encounter phases.

Discussion:

The early rule was: do not train/deploy one universal residual model for all Zone-3 encounter phases. At that time it was not a final theorem, but it was a useful engineering warning. We therefore focused on strong-approach states, because the relevant rollout event was a strong approach.

Decision: Move to velocity-safe training and gate sweeps, with the strong-approach gate as the main candidate.

6.3 Velocity-safe v2 training

Files used / created:

- train_encounter_surrogate_v2_velocitysafe.py
- train_encounter_surrogate_v2_velocitysafe_fixed.py
- train_encounter_surrogate_v2_velocitysafe_fixed2.py

Data, models, and outputs:

- encounter_surrogate_v2_velocitysafe_large1.pt
- encounter_surrogate_v2_velocitysafe_large1_summary.txt
- v2 prediction/evaluation outputs.

Result:

The v2 training objective explicitly weighted velocity safety. During development, code bugs were found and fixed: first a dataset/tensor tuple indexing issue, then a variable-name issue in the loss accumulation path. After fixed2 ran successfully, the model showed improved position behavior but velocity remained insufficiently solved.

Discussion:

This phase clarified that position-only success was not enough. The residual model had to be safe in both position and velocity, because velocity errors at the encounter exit propagate strongly through the rest of the chaotic rollout.

Decision: Do not train again blindly; run gate sweeps and test deployment behavior.

6.4 Gate sweep and first rollout prototype

Files used / created:

- evaluate_encounter_surrogate_gate_sweep_v2.py
- evaluate_encounter_surrogate_fixed_gate_v2.py
- evaluate_gated_encounter_surrogate_rollout_v1.py

Data, models, and outputs:

- encounter_surrogate_v2_velocitysafe_gate_sweep.txt
- encounter_surrogate_v2_velocitysafe_fixed_gate_vr_m0p40.txt
- rollout_surrogate_v1_* folders.

Result:

The gate sweep identified $v_{\text{rad_norm}} < -0.40$ as a plausible strong-approach deployment rule. Initial rollout testing showed either no meaningful event or fallback behavior. Further inspection identified the key real rollout event around $t = 82.56$ yr.

Discussion:

This was the first important deployment lesson: offline test-set performance was not enough. The model had to work on the exact encounter state produced by the long noNN rollout.

Decision: Generate data targeted around the real event geometry: $v_{\text{rad_norm}} \sim -1.1$, $v_{\text{tan_norm}} \sim 0.85$, $r_{\text{pair}} \sim 0.139$ AU, window $\min_r \sim 0.062$ AU, $dt = 0.08$.

6.5 Event-targeted dataset

Files used / created:

- generate_encounter_surrogate_data_v2_event_targeted.py
- generate_encounter_surrogate_data_v2_event_targeted_fixed.py
- merge_encounter_surrogate_shards_v1.py
- inspect_encounter_surrogate_data_v1.py

Data, models, and outputs:

- event_strong_approach shards in encounter_surrogate_shards/
- Merged event-enriched dataset reaching 2870 samples before rollout-local enrichment.
- encounter_surrogate_v2_event_enriched_velocitysafe.pt and associated gate-sweep/fixed-gate outputs.

Result:

The targeted generator produced strong-approach samples around the desired scalar event values. The merged dataset became much more concentrated around the difficult local encounter regime.

Discussion:

This helped but did not fully solve the failure. The exact rollout event was still not sufficiently covered in full 18D state-space. The data looked close in scalar descriptors but was not close enough in the actual state representation used by the model.

Decision: Diagnose full-state coverage explicitly with nearest-neighbor analysis.

6.6 Full-state nearest-neighbor diagnosis

Files used / created:

- diagnose_event_nearest_neighbors_v1.py
- diagnose_knn_residual_surrogate_v1.py
- diagnose_single_event_replay_knn_v1.py

Data, models, and outputs:

- event_nn_diag_t82p56/
- knn_residual_diag_v1/
- single-event KNN replay outputs.

Result:

The nearest-neighbor analysis showed the real $t = 82.56$ event was not well covered in X_{rel18} /full-state space, even though scalar features were similar. KNN-style residuals were therefore not reliable enough to fix the exact event.

Discussion:

This changed the interpretation of the failure: the issue was not simply MLP capacity or scalar gate selection. It was a data-support problem. The model was being asked to interpolate/extrapolate to a full-state geometry that was not sufficiently represented.

Decision: Generate data directly from the actual IC1 noNN rollout geometry, not just from scalar-targeted synthetic states.

6.7 Compact and hybrid representation attempts

Files used / created:

- train_encounter_surrogate_v3_compact.py
- train_encounter_surrogate_v4_hybrid.py

Data, models, and outputs:

- encounter_surrogate_v3_compact_event_enriched_summary.txt
- encounter_surrogate_v4_hybrid_event_enriched_summary.txt.

Result:

The compact representation and hybrid architecture were useful tests but did not solve the exact-event issue. A scalar or reduced representation cannot reliably correct an entire 18D exit-state residual unless the data support and model target are aligned.

Discussion:

These attempts were not wasted: they confirmed that representation tweaks alone were not enough. The exact local rollout state had to be represented in the training distribution.

Decision: Stop architecture-only iteration and generate rollout-local data.

6.8 Rollout-local dataset generation

Files used / created:

- generate_encounter_surrogate_data_v3_rollout_local.py
- merge_encounter_surrogate_shards_v1.py
- inspect_encounter_surrogate_data_v1.py

Data, models, and outputs:

- Rollout-local batches 202-206, 100 samples each.
- encounter_surrogate_v3_rollout_local_enriched.npz
- inspect_surrogate_v3_rollout_local_enriched/

Result:

The rollout-local data directly anchored samples on the actual IC1 revised noNN rollout around $t = 82.56$ yr. After merging, the final training dataset had 3370 samples: 2870 previous enriched samples plus 500 rollout-local samples. Inspection found no red flags.

Discussion:

This was the decisive data improvement. It placed training examples near the actual failed event in full-state geometry and residual scale, rather than only in scalar feature space.

Decision: Train the final velocity-safe full-state residual model on the rollout-local enriched dataset.

6.9 Final velocity-safe rollout-local model

Files used / created:

- train_encounter_surrogate_v2_velocitysafe_fixed2.py

Data, models, and outputs:

- encounter_surrogate_v3_rollout_local_velocitysafe.pt
- encounter_surrogate_v3_rollout_local_velocitysafe_summary.txt

Result:

The final model showed strong held-out gains. Overall, position median improved from about $8.43\text{e-}03$ to $1.28\text{e-}03$ and velocity median improved from about $2.62\text{e-}02$ to $5.05\text{e-}03$. On strong-approach test samples, position median improved by about 86.9% and velocity median by about 89.0%.

Discussion:

The model was now good enough for exact-event replay. The remaining concern was unsafe tails, so it still required gated deployment rather than blind use everywhere.

Decision: Test the exact real event at $t = 82.56$ yr.

6.10 Exact single-event replay

Files used / created:

- diagnose_single_event_replay_v1.py

Data, models, and outputs:

- single_event_replay_v3_rollout_local_t82p56/single_event_replay_summary.txt
- single_event_replay_v3_rollout_local_t82p56/single_event_replay_metrics.csv

Result:

At the exact event, the true residual RMS was position $6.27095\text{e-}03$ and velocity $2.85511\text{e-}02$. The predicted residual RMS was position $7.23297\text{e-}03$ and velocity $2.74930\text{e-}02$. After applying the correction, local 0.5-year exit error improved to position $1.39569\text{e-}03$ and velocity $1.39594\text{e-}03$, corresponding to gains of 77.74% and 95.11%.

Discussion:

This proved the local encounter map was learnable once rollout-local data were added. The original overcorrection failure was fixed locally.

Decision: Proceed to full 100-year rollout to test whether local accuracy improves global metrics.

6.11 Full rollout and alpha timeline diagnostics

Files used / created:

- evaluate_gated_encounter_surrogate_rollout_v1.py
- diagnose_rollout_error_timeline_v1.py
- diagnose_rollout_error_timeline_v1b.py
- diagnose_rollout_error_timeline_v1c.py

Data, models, and outputs:

- rollout_surrogate_v3_rollout_local_T100_vr_m0p40/
- rollout_error_timeline_v1c/

Result:

The full rollout used the encounterNN once at $t = 82.56$ yr. It improved final position and velocity, and improved velocity time-average, but slightly worsened position time-average. The alpha sweep showed that partial correction did not solve the branch problem; $\alpha=1.0$ was the best of the tested values.

Discussion:

This revealed the second major limitation: even a locally better exit state can lead to a different chaotic branch over the remaining years. The issue was not simply correction magnitude.

Decision: Use the full correction in the final production-like evaluator, but compare honestly against scalarNN and noNN.

7. Final Production-Like Ablation Evaluator

The final evaluator was built as `pair_eval_encounterNN_ablation_v1.py`. Its purpose was to put encounterNN into the same style of full-run comparison as the previous scalar Zone-3 evaluator. It compares four methods on IC1 at $dt = 0.08$, $T = 100$ yr, $n_{\text{samples}} = 5000$:

- IAS15 reference.
- SIMON-noNN: revised noNN Zone-2/Zone-3 path.
- SIMON-scalarNN: existing bounded v4 scalar Zone-3 model.
- SIMON-encounterNN: noNN rollout plus gated encounter residual correction at strong-approach windows.

The final output files were `encounterNN_ablation_v1_out/encounterNN_ablation_summary.txt` and `encounterNN_ablation_v1_out/encounter_events.csv`.

Method	pos final	pos timeavg	pos p95	vel final	vel timeavg	runtime (s)	speedup vs IAS15
SIMON-noNN	1.6236	0.9891	1.6942	1.1826	0.8650	0.02361	2.96x
SIMON-scalarNN	1.2258	0.9679	1.6427	1.0392	0.8822	0.02534	2.76x
SIMON-encounterNN	1.4302	1.0088	1.7442	0.6291	0.8366	0.13730	0.51x

Delta versus SIMON-noNN:

Method	Position final gain	Position timeavg gain	Velocity final gain	Velocity timeavg gain
SIMON-scalarNN	+24.50%	+2.14%	+12.13%	-1.99%
SIMON-encounterNN	+11.91%	-2.00%	+46.81%	+3.29%

Final event record:

t_start	t_exit	r_pair	v_rad_norm	v_tan_norm	min_r_window	pred_pos_norm	pred_vel_norm	relE_corr
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82.56	83.06	0.138965	-1.114827	0.849319	0.062263	0.007233	0.027493	0.000657
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8. Why the Final Results Were Poor for the Full-Pipeline Goal

1. The encounterNN fired only once in the 100-year IC1 rollout. A one-time state correction is unlikely to outperform a scalar force correction that acts repeatedly during all eligible Zone-3 force evaluations.
2. The local correction worked, but it changed the long-horizon chaotic branch. This improved velocity metrics and final position, but worsened the main position time-average metric.
3. The residual method still runs the 0.5-year noNN window to compute $F_{S,new}$. Therefore it adds work instead of skipping work. This is why encounterNN was much slower in the final evaluator.
4. The scalarNN is better matched to the current SIMON design: it preserves analytic direction, acts inside the existing force computation, and retains the optimized leapfrog/substep structure.
5. The encounterNN model was validated only for $dt = 0.08$, IC1, active pair 1-2, and $window_years = 0.5$. It is not yet a general replacement for the production scalarNN method.

9. Final Conclusions for Mentor Discussion

Conclusion	Evidence
The encounterNN concept is locally valid.	Exact event replay reduced local position exit error by 77.74% and local velocity exit error by 95.11%.
Data coverage was the critical obstacle.	Scalar-targeted data did not cover the exact event in full 18D state-space; rollout-local data fixed the local failure.
Global deployment remains branch-sensitive.	Full rollout position time-average worsened despite local exit improvement; alpha sweep did not remove the branch effect.
The current residual encounterNN is not a speedup method.	It needs the noNN window $F_{S,new}$, then adds NN inference. Final speedup was only 0.51x vs IAS15.
ScalarNN remains the main production method.	ScalarNN achieved better position final and position timeavg metrics, while preserving 2.76x speedup in the final ablation.
EncounterNN belongs in future work or appendix.	It is strong evidence that close-encounter residual maps are learnable, but not yet a replacement for scalar Zone-3 correction.

10. Recommended Paper Framing

Recommended wording:

We tested an encounter-level residual surrogate as a prototype extension to SIMON. After rollout-local data enrichment, the model accurately corrected the difficult IC1 close encounter at $t = 82.56$ yr, reducing local position and velocity exit errors by 77.7% and 95.1%. However, when inserted into the optimized 100-year SIMON pipeline, the correction improved velocity metrics but did not outperform the scalar Zone-3 NN on position time-average or runtime. This suggests that encounter-exit residual maps are learnable, but the current residual-overlay deployment remains branch-sensitive and is not yet a production replacement for scalar force correction.

Recommended placement:

- Main paper: keep existing scalarNN results as the primary SIMON architecture and computational-performance claim.
- Appendix or future work: include encounterNN as a prototype showing that local encounter residual learning can work if data are generated from the actual rollout geometry.
- Do not claim encounterNN speedup or replacement of scalarNN based on the current residual model.

11. Future Work If We Continue EncounterNN

Future direction	Why needed
Direct-exit / skip-window model	To get speedup, the model must avoid running the noNN 0.5-year

	window. A direct-exit model would use a different target and stronger physical constraints.
Broader rollout-local datasets	Current model is validated only around IC1, pair 1-2, dt = 0.08, window = 0.5 yr. A usable method needs many ICs, many encounter geometries, and multiple dt values.
Physics-constrained exit correction	State-level corrections should preserve center of mass, momentum, and energy more structurally rather than relying only on post-checks.
Compiled/vectorized implementation	Python-level residual deployment will not be a fair speed benchmark. A production version needs low-overhead integration with the hot loop.
Branch-aware evaluation	For chaotic systems, future metrics should separate local exit accuracy, final-state accuracy, time-average RMS, boundedness, and phase/branch sensitivity.

12. File and Artifact Inventory

Category	Files / folders	Role
Original scalar baseline	pair_eval_v4_zone3_frozen_timeavg_v3_with_log.py; train_pair_correction_v4.py; pair_correction_nn_v4_bounded.pt; encounter_data_zone3_v3_augmented.npz	Production scalar Zone-3 baseline and training artifacts.
Initial encounter data	generate_encounter_surrogate_data_v1.py; merge_encounter_surrogate_shards_v1.py; inspect_encounter_surrogate_data_v1.py	Generate, merge, and inspect initial encounter-window residual datasets.
Velocity-safe models	train_encounter_surrogate_v2_velocitysafe.py; train_encounter_surrogate_v2_velocitysafe_fixed.py; train_encounter_surrogate_v2_velocitysafe_fixed2.py	Train velocity-aware 18D residual models; fixed code bugs during this phase.
Gate and rollout diagnostics	evaluate_encounter_surrogate_gate_sweep_v2.py; evaluate_encounter_surrogate_fixed_gate_v2.py; evaluate_gated_encounter_surrogate_rollout_v1.py	Gate selection and full rollout testing.
Event-targeted data	generate_encounter_surrogate_data_v2_event_targeted.py; generate_encounter_surrogate_data_v2_event_targeted_fixed.py	Targeted strong-approach data around scalar features of the real event.
Coverage diagnostics	diagnose_event_nearest_neighbors_v1.py; diagnose_knn_residual_surrogate_v1.py; diagnose_single_event_replay_knn_v1.py	Showed that scalar coverage was insufficient in full-state X_rel18 space.
Representation tests	train_encounter_surrogate_v3_compact.py; train_encounter_surrogate_v4_hybrid.py	Tested compact and hybrid alternatives; did not solve the exact event problem.
Rollout-local data	generate_encounter_surrogate_data_v3_rollout_local.py; encounter_surrogate_v3_rollout_local_enriched.npz; inspect_surrogate_v3_rollout_local_enriched/	Final successful dataset source; 3370 samples after adding 500 rollout-local samples.
Final encounter model	encounter_surrogate_v3_rollout_local_velocitysafe.pt; encounter_surrogate_v3_rollout_local_velocitysafe_summary.txt	Final trained 18D residual model.
Exact event / alpha diagnostics	diagnose_single_event_replay_v1.py; diagnose_rollout_error_timeline_v1c.py; single_event_replay_v3_rollout_local_t82p56/; rollout_error_timeline_v1c/	Validated local correction and diagnosed branch sensitivity.
Final evaluator	pair_eval_encounterNN_ablation_v1.py; encounterNN_ablation_v1_out/encounterNN_ablation_summary.txt; encounterNN_ablation_v1_out/encounter_events.csv	Final full-pipeline ablation against noNN and scalarNN.

Appendix A. Final Ablation Output

The final evaluator recorded the following raw summary values:

SIMON encounterNN ablation v1

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Purpose: compare IAS15, SIMON-noNN, SIMON-scalarNN, and SIMON-encounterNN on IC1.

EncounterNN deployment formula: corrected_exit = noNN_exit + predicted_residual.

The noNN and scalarNN baselines are run through the original v4 evaluator path.

scalar_model: C:\Aarush\Physics\training\encounter_training\pair_correction_nn_v4_bounded.pt

encounter_model: encounter_surrogate_v3_rollout_local_velocitysafe.pt

dt: 0.08000000

T_years: 100.00000000

n_samples: 5000

window_years: 0.50000000

vr_thresh: -0.40000000

energy_gate: 0.20000000

device: cuda

Runtime

IAS15_total_time_sec: 0.069805500

SIMON_noNN_total_time_sec: 0.023610700

SIMON_noNN_speedup_vs_IAS15: 2.956520

SIMON_scalarNN_total_time_sec: 0.025337300

SIMON_scalarNN_speedup_vs_IAS15: 2.755049

SIMON_encounterNN_total_time_sec: 0.137298100

SIMON_encounterNN_speedup_vs_IAS15: 0.508423

Accuracy vs IAS15

method	pos_final total_time_sec	pos_timeavg speedup_vs_ias15	pos_med	pos_p95	vel_final	vel_timeavg	vel_med	vel_p95
SIMON-noNN	1.623558607e+00 4.549183358e-01	9.890699386e-01 1.539884282e+00	6.703518724e-01 0.023610700		1.694171640e+00 2.956520	1.182585762e+00	8.650211065e-01	
SIMON-scalarNN	1.225807191e+00 4.549183358e-01	9.679291352e-01 1.540645134e+00	6.703518724e-01 0.025337300		1.642698716e+00 2.755049	1.039164176e+00	8.822249710e-01	
SIMON-encounterNN	1.430212968e+00 8.365985364e-01	1.008840117e+00 4.549183358e-01	6.703518724e-01 1.539884282e+00		1.744244643e+00 0.137298100	1.744244643e+00 0.508423	6.290521326e-01	

EncounterNN event stats

gate_candidates: 3
encounter_attempts: 1
encounter_used: 1
encounter_fallback: 0
first_used_t: 82.5599999999986
last_used_t: 82.5599999999986
pred_pos_norm_mean: 0.007232969460293762
pred_vel_norm_mean: 0.027493012715130775
min_r: 0.06226251907993296
max_radius: 1.9359932171802954
event_csv: encounterNN_ablation_v1_out\encounter_events.csv

Delta vs SIMON-noNN

SIMON-encounterNN_pos_final_gain_pct: +11.908756
SIMON-encounterNN_pos_timeavg_gain_pct: -1.998866
SIMON-encounterNN_vel_final_gain_pct: +46.807060
SIMON-encounterNN_vel_timeavg_gain_pct: +3.285766
SIMON-scalarNN_pos_final_gain_pct: +24.498741
SIMON-scalarNN_pos_timeavg_gain_pct: +2.137443
SIMON-scalarNN_vel_final_gain_pct: +12.127796
SIMON-scalarNN_vel_timeavg_gain_pct: -1.988838

Notes

- noNN and scalarNN use the original v4 Zone2/Zone3 evaluator path, so their accuracy should match the v3/v4 log file for the same model, dt, T, n_samples, and environment.
- Runtime can vary slightly across runs and hardware; numerical accuracy values should match up to normal floating-point/order effects.
- EncounterNN is currently validated for dt=0.08, IC1, active pair 1-2, window_years=0.5.